# Thrust-to-weight ratio

**Thrust-to-weight ratio** is a dimensionless ratio of thrust to weight of a reaction engine or a vehicle with such an engine. Reaction engines include jet engines, rocket engines, pump-jets, Hall-effect thrusters, and ion thrusters, among others. These generate thrust by expelling mass (propellant) in the opposite direction of intended motion, in accordance with Newton's third law. A related but distinct metric is the power-to-weight ratio, which applies to engines or systems that deliver mechanical, electrical, or other forms of power rather than direct thrust.

In many applications, the thrust-to-weight ratio serves as an indicator of performance. The ratio in a vehicle's initial state is often cited as a figure of merit, enabling quantitative comparison across different vehicles or engine designs. The instantaneous thrust-to-weight ratio of a vehicle can vary during operation due to factors such as fuel consumption (which reduces mass) or changes in gravitational acceleration, for example in orbital or interplanetary contexts.

## Calculation

Thrust-to-weight from mass (assuming standard gravity)

| | |
|---|---|
| Thrust | 1500 N |
| Mass | 17.1 kg |
| TWR | **~8.945** |

The thrust-to-weight ratio of an engine or vehicle is calculated by dividing its thrust by its weight (not to be confused with mass). The formula is:

$$\mathrm{TWR} = \frac{T}{W} = \frac{T}{m \cdot g}$$

where:

- $T$ is the **thrust**, in newtons (N), kilograms-force (kgf), or pounds-force (lbf),
- $W$ is the **weight**, in newtons (N), which can also be expressed as the product of:
  - **mass** $m$, in kilograms (kg) or pounds (lb), and
  - **gravitational acceleration** $g$, e.g., the standard gravitational acceleration on Earth of 9.80665 m/s$^2$.

For valid comparison of the initial thrust-to-weight ratio of two or more engines or vehicles, thrust must be measured under controlled conditions. Because an aircraft's weight can vary considerably, depending on factors such as munition load, fuel load, cargo weight, or even the weight of the pilot, the thrust-to-weight ratio is also variable and even changes during flight operations. There are several standards for determining the weight of an aircraft used to calculate the thrust-to-weight ratio range.

- **Empty weight** – The weight of the aircraft minus fuel, munitions, cargo, and crew.
- **Combat weight** – Primarily for determining the performance capabilities of fighter aircraft, it is the weight of the aircraft with full munitions and missiles, half fuel, and no drop tanks or bombs.
- **Max takeoff weight** – The weight of the aircraft when fully loaded with the maximum fuel and cargo that it can safely takeoff with.[1]

## Aircraft

The thrust-to-weight ratio and lift-to-drag ratio are the two most important parameters in determining the performance of an aircraft.

The thrust-to-weight ratio varies continually during a flight. Thrust varies with throttle setting, airspeed, altitude, air temperature, etc. Weight varies with fuel burn and payload changes. For aircraft, the quoted thrust-to-weight ratio is often the maximum static thrust at sea level divided by the maximum takeoff weight.[2] Aircraft with thrust-to-weight ratio greater than 1:1 can pitch straight up and maintain airspeed until performance decreases at higher altitude.[3]

A plane can take off even if the thrust is less than its weight as, unlike a rocket, the lifting force is produced by lift from the wings, not directly by thrust from the engine. As long as the aircraft can produce enough thrust to travel at a horizontal speed above its stall speed, the wings will produce enough lift to counter the weight of the aircraft.

$$\left(\frac{T}{W}\right)_{\text{cruise}} = \left(\frac{D}{L}\right)_{\text{cruise}} = \frac{1}{\left(\frac{L}{D}\right)_{\text{cruise}}}.$$

### Propeller-driven aircraft

For propeller-driven aircraft, the thrust-to-weight ratio can be calculated as follows in imperial units:[4]

$$\frac{T}{W} = \frac{550\eta_{\mathrm{p}}}{V}\frac{\text{hp}}{W},$$

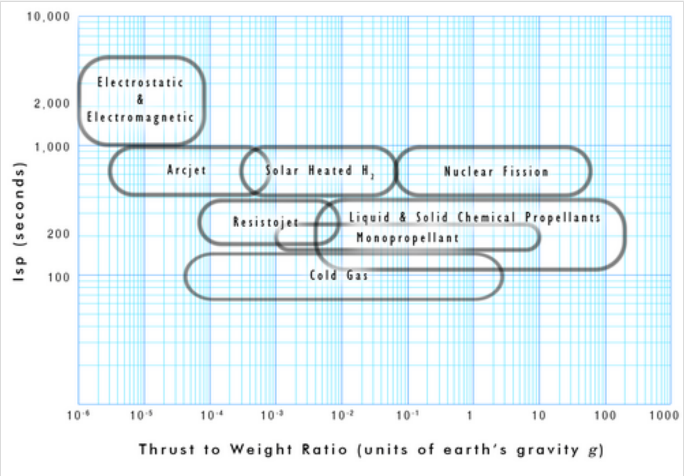
where $\eta_{\mathrm{p}}$ is propulsive efficiency (typically 0.65 for wooden propellers, 0.75 metal fixed pitch and up to 0.85 for constant-speed propellers), hp is the engine's shaft horsepower, and $V$ is true airspeed in feet per second, weight is in lbs.

The metric formula is:

$$\frac{T}{W} = \left(\frac{\eta_{\mathrm{p}}}{V}\right)\left(\frac{P}{W}\right).$$

## Rockets

The thrust-to-weight ratio of a rocket, or rocket-propelled vehicle, is an indicator of its acceleration expressed in multiples of gravitational acceleration *g*.[5]



Rocket vehicle thrust-to-weight ratio vs specific impulse for different propellant technologies

Rockets and rocket-propelled vehicles operate in a wide range of gravitational environments, including the *weightless* environment. The thrust-to-weight ratio is usually calculated from initial gross weight at sea level on earth[6] and is sometimes called *thrust-to-Earth-weight ratio*.[7] The thrust-to-Earth-weight ratio of a rocket or rocket-propelled vehicle is an indicator of its acceleration expressed in multiples of earth's gravitational acceleration, $g_0$.[5]

The thrust-to-weight ratio of a rocket improves as the propellant is burned. With constant thrust, the maximum ratio (maximum acceleration of the vehicle) is achieved just before the propellant is fully consumed. Each rocket has a characteristic thrust-to-weight curve, or acceleration curve, not just a scalar quantity.

The thrust-to-weight ratio of an engine is greater than that of the complete launch vehicle, but is nonetheless useful because it determines the maximum acceleration that *any* vehicle using that engine could theoretically achieve with minimum propellant and structure attached.

For a takeoff from the surface of the earth using thrust and no aerodynamic lift, the thrust-to-weight ratio for the whole vehicle must be greater than *one*. In general, the thrust-to-weight ratio is numerically equal to the *g-force* that the vehicle can generate.[5] Take-off can occur when the vehicle's *g-force* exceeds local gravity (expressed as a multiple of $g_0$).

The thrust-to-weight ratio of rockets typically greatly exceeds that of airbreathing jet engines because the comparatively far greater density of rocket fuel eliminates the need for much engineering materials to pressurize it.

Many factors affect thrust-to-weight ratio. The instantaneous value typically varies over the duration of flight with the variations in thrust due to speed and altitude, together with changes in weight due to the amount of remaining propellant, and payload mass. Factors with the greatest effect include freestream air temperature, pressure, density, and composition. Depending on the engine or vehicle under consideration, the actual performance will often be affected by buoyancy and local gravitational field strength.

# Examples

## Aircraft

| Vehicle | thrust-weight ratio | Notes |
|---|---|---|
| Northrop Grumman B-2 Spirit | 0.205[8] | Max take-off weight, full power |
| Airbus A340-300 Enhanced | 0.2229 | Max take-off weight, full power |
| Airbus A380 | 0.227 | Max take-off weight, full power |
| Boeing 747-8 | 0.269 | Max take-off weight, full power |
| Boeing 777-200ER | 0.285 | Max take-off weight, full power |
| Boeing 737 MAX 8 | 0.311 | Max take-off weight, full power |
| Airbus A320neo | 0.310 | Max take-off weight, full power |
| Boeing 757-200 | 0.341 | Max take-off weight, full power (w/Rolls-Royce RB211) |
| Tupolev 154B | 0.360 | Max take-off weight, full power (w/Kuznetsov NK-8-2) |
| Tupolev Tu-160 | 0.363 | Max take-off weight, full afterburners |
| Concorde | 0.372 | Max take-off weight, full afterburners |
| Rockwell International B-1 Lancer | 0.38 | Max take-off weight, full afterburners |
| HESA Kowsar | 0.61 | With full fuel, afterburners. |
| BAE Hawk | 0.65[9] | |
| Lightning F.6 | 0.702 | Max take-off weight, full afterburners |
| Lockheed Martin F-35 A | 0.87 | With full fuel (1.07 with 50% fuel, 1.19 with 25% fuel) |
| HAL Tejas Mk 1 | 1.07 | With full fuel |
| CAC/PAC JF-17 Thunder | 1.07 | With full fuel |
| Dassault Rafale | 1.028 (1.219 with loaded weight & 50% internal fuel) | Version C, 100% fuel |
| Sukhoi Su-30MKM | 1.00[10] | Loaded weight with 56% internal fuel |
| McDonnell Douglas F-15 | 1.04[11] | Nominally loaded |
| Mikoyan MiG-29 | 1.09[12] | Full internal fuel, 4 AAMs |
| Lockheed Martin F-22 | >1.09 (1.26 with loaded weight and 50% fuel)[13] | |
| General Dynamics F-16 | 1.096 (1.24 with loaded weight & 50% fuel) | |
| Hawker Siddeley Harrier | 1.1 | VTOL |
| Eurofighter Typhoon | 1.15[14] | Interceptor configuration |
| Space Shuttle | 1.3[15] | Take-off |
| Simorgh (rocket) | 1.83 | |
| Space Shuttle | 3 | Peak |

## Jet and rocket engines

| Engine | Mass | Thrust, vacuum | | Thrust-to-weight ratio |
|---|---|---|---|---|
| | | (kN) | (lbf) | |
| MD-TJ42 powered sailplane jet engine[16] | 3.85kg (8.48 lb) | 0.35 | 78.7 | 9.09 |
| RD-0410 nuclear rocket engine[17][18] | 2,000 kg (4,400 lb) | 35.2 | 7,900 | 1.8 |
| Pratt & Whitney J58 jet engine (Lockheed SR-71 Blackbird)[19][20] | 2,722 kg (6,001 lb) | 150 | 34,000 | 5.6 |
| Rolls-Royce/Snecma Olympus 593 turbojet with reheat (Concorde)[21] | 3,175 kg (7,000 lb) | 169.2 | 38,000 | 5.4 |
| Williams FJ33-5A | 140 kg (310 lb) | 8.21 | 1,846 | 5.98 |
| Pratt & Whitney F119[22] | 1,800 kg (4,000 lb) | 91 | 20,500 | 7.95 |
| PBS TJ40-G1NS jet engine[23] | 3.6 kg (7.9 lb) | 0.425 | 96 | 12.04 |
| RD-0750 rocket engine three-propellant mode[24] | 4,621 kg (10,188 lb) | 1,413 | 318,000 | 31.2 |
| RD-0146 rocket engine[25] | 260 kg (570 lb) | 98 | 22,000 | 38.4 |
| Rocketdyne RS-25 rocket engine (Space Shuttle Main Engine)[26] | 3,177 kg (7,004 lb) | 2,278 | 512,000 | 73.1 |
| RD-180 rocket engine[27] | 5,393 kg (11,890 lb) | 4,152 | 933,000 | 78.7 |
| RD-170 rocket engine | 9,750 kg (21,500 lb) | 7,887 | 1,773,000 | 82.5 |
| F-1 (Saturn V first stage)[28] | 8,391 kg (18,499 lb) | 7,740.5 | 1,740,100 | 94.1 |
| NK-33 rocket engine[29] | 1,222 kg (2,694 lb) | 1,638 | 368,000 | 136.7 |
| SpaceX Raptor 3 rocket engine[30] | 1,525 kg (3,362 lb) | 2,746 | 617,000 | 183.6 |
| Merlin 1D rocket engine, full-thrust version[31][32] | 467 kg (1,030 lb) | 914 | 205,500 | 199.5 |
| Williams F107 | 30 kg (66 lb) | 1.9 | 430 | 6.46 |

## Fighter aircraft

Thrust-to-weight ratios, fuel weights, and weights of different fighter planes

| Specifications | F-15K[a] | F-15C | MiG-29K | MiG-29B | JF-17 | J-10 | F-35A | F-35B | F-35C | F-22 | LCA Mk-1 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Engines thrust, maximum (N) | 259,420 (2) | 208,622 (2) | 176,514 (2) | 162,805 (2) | 84,400 (1) | 122,580 (1) | 177,484 (1) | 177,484 (1) | 177,484 (1) | 311,376 (2) | 84,516 (1) |
| Aircraft mass, empty (kg) | 17,010 | 14,379 | 12,723 | 10,900 | 7,965 | 09,250 | 13,290 | 14,515 | 15,785 | 19,673 | 6,560 |
| Aircraft mass, full fuel (kg) | 23,143 | 20,671 | 17,963 | 14,405 | 11,365 | 13,044 | 21,672 | 20,867 | 24,403 | 27,836 | 9,500 |
| Aircraft mass, max. take-off load (kg) | 36,741 | 30,845 | 22,400 | 18,500 | 13,500 | 19,277 | 31,752 | 27,216 | 31,752 | 37,869 | 13,500 |
| Total fuel mass (kg) | 06,133 | 06,292 | 05,240 | 03,505 | 02,300 | 03,794 | 08,382 | 06,352 | 08,618 | 08,163 | 02,458 |
| T/W ratio, full fuel | 1.14 | 1.03 | 1.00 | 1.15 | 1.07 | 1.05 | 0.84 | 0.87 | 0.74 | 1.14 | 1.07 |
| T/W ratio, max. take-off load | 0.72 | 0.69 | 0.80 | 0.89 | 0.70 | 0.80 | 0.57 | 0.67 | 0.57 | 0.84 | 0.80 |

- Table for Jet and rocket engines: jet thrust is at sea level

- Fuel density used in calculations: 0.803 kg/l
- For the metric table, the *T*/*W* ratio is calculated by dividing the thrust by the product of the full fuel aircraft weight and the acceleration of gravity.
- J-10's engine rating is of AL-31FN.

## See also

- Power-to-weight ratio
- Factor of safety

## Notes

a. Pratt & Whitney engines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4. Daniel P. Raymer, *Aircraft Design: A Conceptual Approach*, Equations 3.9 and 5.1
5. George P. Sutton & Oscar Biblarz, *Rocket Propulsion Elements* (p. 442, 7th edition) "thrust-to-weight ratio $F/W_g$ is a dimensionless parameter that is identical to the acceleration of the rocket propulsion system (expressed in multiples of $g_0$) if it could fly by itself in a gravity-free vacuum"
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## External links

- NASA webpage with overview and explanatory diagram of aircraft thrust to weight ratio (https://www.grc.nasa.gov/WWW/K-12/airplane/fwrat.html)